

Adaptive Behavior and Objectives

EES 4760/5760

Agent-Based and Individual-Based Computational Modeling

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Announcements

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- Homework assigned for Wednesday, October 7 is **optional**.
 - There will be no more required homework exercises from the textbook except the exercises for your team's team project.
 - From here on, focus on working on your team project and individual project.
- On Friday, I posted the exercises your team should work on for the team project
 - Each team has a problem to work on for your in-class presentation and several other problems to work on for your written report.
 - There is one problem for each person in your team plus the one for presentation
 - You can work on all the problems together, or have each team member work on a different problem, and work together on the presentation problem
 - Your written report should include your answers to all the problems (including the presentation one).

Sensing

Sensing

- Options for sensing:
 - Omnisicence: `max-one-of [ expected-utility ] patches`
 - Neighbors: `max-one-of [ expected-utility ] neighbors`
 - Limited radius: `max-one-of [expected-utility ] patches in-radius 5`
 - Social network: `max-one-of [ expected-utility ] [neighbors] of out-link-neighbors`
- Context:
 - NetLogo has **four** types of entities:
 1. Patches
 2. Turtles
 3. Links
 4. The Observer

Social Networks and Links

- Links
 - Connect turtles
 - Directed (`create-link-from`, `create-link-to`) or undirected (`create-link-with`)
 - Can have properties (color, thickness, etc.)
- Using links:
 - `my-links`, `my-in-links`, `my-out-links` report agent-sets of **links** connected to a turtle
 - `link-neighbors`, `out-link-neighbors`, `in-link-neighbors` report agent-sets of **turtles** connected to a turtle.
 - **Lots** more you can do with links (read NetLogo dictionary)

Adaptation

Adaptation and Objectives

- Making decisions:
 - Perfect rationality:
 - Pick a goal (objective function)
 - List possible actions
 - Calculate how well each will satisfy goal
 - Choose action that will best accomplish goal
 - Imperfect rationality:
 - Goal may be unclear or inconsistent
 - May not list all possible actions
 - May not calculate results of actions
 - May not act on best option
- Real-life agents may not act rationally

Bounded Rationality

- Perfect rationality and chess ...
 - Evaluating all possible moves may not be possible
 - Limited time, memory, computing power
 - Cost of rationality
 - Getting, processing information
 - It may be more rational to be slightly irrational

Satisficing

- Define goal (objective function)
- Define criteria for *good enough* result
- Evaluate possible actions until the first one that is *good enough*.
 - Do that action.

Business-Investor Satisficing

- The investor can't see the utility of any other patches
- If the utility of the current patch is greater than a certain threshold, then it's **“good enough”**
 - If the current patch isn't good enough:
 - Move to a random empty patch
 - neighbor, in-radius, link neighbor, etc.

More on Objective Functions

- Decisions under uncertainty
 - If you are gambling, what would you try to do?
 - Take a chance to get a very big win?
 - Try to avoid losing money?
 - Balance wins and losses finish with the most money on average?
- Behavioral economics
 - Most economists say rational people will try to get the greatest *expected* wealth
 - Actual people may be...
 - *risk seeking* (take greater chances for big wins)
 - *risk averse* (avoid taking chances)
 - *loss averse* (focus on changes instead of absolute wealth)
 - *regret averse* (try to avoid the feeling that you wish you'd made a different choice)
- Different goals may lead to very different behavior
 - Policy-makers may want to test their policies under different assumptions about people's goals and behavior

Experimenting

Experimenting

- Modified Business Investor Model:
 - Vary the profits and risks
 - Vary the sensing:
 - Neighboring patches
 - Patches in a radius
 - Neighbors or patches in-radius of link neighbors
 - Vary the objective and adaptation
 - Maximize expected utility
 - Satisficing